Evidencia de ejecución

Integrantes: David Domínguez (23712), Jonathan Diaz (23873) y Luis Padilla (23663)
Repositorio: https://github.com/Jonialen/labPCAM

Resultados principales

C11: secuencial 0.8946 s; directa 8/static 0.2048 s (4.37×); por pares 8/static 0.2340 s. C++17: secuencial 0.8892 s; directa 8/static 0.2062 s (4.31×); por pares 8/static 0.2396 s. Los valores son medias de tres ejecuciones con N=5000, 10 pasos y semilla 42.

La versión directa aceleró el cálculo respecto a la secuencial. La versión por pares realizó menos interacciones, pero tardó más que la directa de 8 hilos con el mismo schedule static: sus acumuladores privados y la reducción agregan trabajo.

Las capturas son ejecuciones adicionales con la misma carga y semilla; sus tiempos individuales pueden diferir de las medias del CSV.

1. Compilación exitosa

```
$ wsl.exe -d Ubuntu-24.04 -- make -s -B all
Compilación completada. Ejecutables generados:
-rwxrwxrwx 1 dom dom 23K Oct 10  2026 build/cpp_nbody_optimized
-rwxrwxrwx 1 dom dom 22K Oct 10 22:03 build/cpp_nbody_parallel
-rwxrwxrwx 1 dom dom 18K Oct 10 22:03 build/cpp_nbody_seq
-rwxrwxrwx 1 dom dom 22K Oct 10 22:03 build/nbody_optimized
-rwxrwxrwx 1 dom dom 22K Oct 10 22:03 build/nbody_parallel
-rwxrwxrwx 1 dom dom 18K Oct 10 22:03 build/nbody_seq
Exit code: 0
PS C:\Users\David\Documents\2026\S2\PARALELA\labPCAM>
```

2. Ejecución secuencial

```
$ wsl.exe -d Ubuntu-24.04 -- build/nbody_seq 5000 10
Students: David Dominguez (23712), Jonathan Diaz (23873), Luis Padilla (23663)
Mode: sequential
N: 5000 | Steps: 10 | Seed: 42
Elapsed: 0.926782674 s
Checksum: 4416.553277470315
Processors reported by OpenMP: 12
Maximum threads reported by OpenMP: 12
Exit code: 0
PS C:\Users\David\Documents\2026\S2\PARALELA\labPCAM>
```

3. Ejecución paralela directa

```
$ wsl.exe -d Ubuntu-24.04 -- build/nbody_parallel 5000 10 8 static
Students: David Dominguez (23712), Jonathan Diaz (23873), Luis Padilla (23663)
Mode: parallel-direct
N: 5000 | Steps: 10 | Seed: 42
Threads requested: 8 | Threads used: 8
Schedule: static | Chunk: default
Elapsed: 0.220668811 s
Checksum: 4416.553277470315
Processors reported by OpenMP: 12
Maximum threads reported by OpenMP: 12
Exit code: 0
PS C:\Users\David\Documents\2026\S2\PARALELA\labPCAM>
```

4. Ejecución optimizada por pares

```
$ wsl.exe -d Ubuntu-24.04 -- build/nbody_optimized 5000 10 8 static
Students: David Dominguez (23712), Jonathan Diaz (23873), Luis Padilla (23663)
Mode: parallel-optimized
N: 5000 | Steps: 10 | Seed: 42
Threads requested: 8 | Threads used: 8
Schedule: static | Chunk: default
Elapsed: 0.257507841 s
Checksum: 4416.553277470315
Processors reported by OpenMP: 12
Maximum threads reported by OpenMP: 12
Exit code: 0
PS C:\Users\David\Documents\2026\S2\PARALELA\labPCAM>
```

5. Hardware y compiladores

```
$ wsl.exe -d Ubuntu-24.04 -- lscpu

Architecture:                         x86_64
CPU(s):                               12
Model name:                           AMD Ryzen 5 4600H with Radeon Graphics
Thread(s) per core:                   2
Core(s) per socket:                   6
Socket(s):                            1
$ wsl.exe -d Ubuntu-24.04 -- gcc -dumpfullversion
13.3.0
$ wsl.exe -d Ubuntu-24.04 -- g++ -dumpfullversion
13.3.0
Exit code: 0


PS C:\Users\David\Documents\2026\S2\PARALELA\labPCAM>
```

6. Comparación del estado completo

```
$ wsl.exe -d Ubuntu-24.04 -- build/compare_full_state 5000 10
N=5000 steps=10 seed=42 direct_threads=8 optimized_threads=8
direct_max_abs_state_difference=0
optimized_max_abs_state_difference=1.0658141036401503e-13
direct_checksum_difference=0
optimized_checksum_difference=0
$ wsl.exe -d Ubuntu-24.04 -- build/cpp_compare_full_state 5000 10
N=5000 steps=10 seed=42 direct_threads=8 optimized_threads=8
direct_max_abs_state_difference=0
optimized_max_abs_state_difference=1.0658141036401503e-13
direct_checksum_difference=0
optimized_checksum_difference=0
Exit code: 0
PS C:\Users\David\Documents\2026\S2\PARALELA\labPCAM>
```